

INFOE 3 - LAB2_Data_Swapping

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BSINFO 3C

The screenshot displays a Visual Studio Code (VS Code) environment with a dark theme. The Explorer panel on the left shows a project named 'LAB2_INFOE3_FRANCISCO_MANULAT_LABANON_CORTEJO_3C' containing files: 'employee_data.csv', 'lab2.py', 'swapped_employee_data.csv', and 'swapping_lab2.py'. The main editor window is open to 'lab2.py', which contains the following Python code:

```
1 import random
2 import pandas as pd
3 from pprint import pprint
4 from faker import Faker
5
6 # Initialize the Faker generator
7 fake = Faker('en_US')
8
9
10 def generate_raw_data(num_entries=25):
11     raw_data = []
12     for _ in range(num_entries):
13         entry = {
14             "name": fake.name(),
15             "age": random.randint(18, 85),
16             "zip_code": fake.postcode(),
17             "salary": random.randint(35000, 150000),
18             "medical_risk": random.choice(["Low", "Medium", "High"])
19         }
20         raw_data.append(entry)
21     return raw_data
22
23 # Execute and print
24 data = generate_raw_data(25)
```

Below the code editor, the TERMINAL panel shows the output of running 'python lab2.py' in a PowerShell window. The output indicates that 25 entries were generated and the dataset was saved to 'employee_data.csv'. A sample of the generated data is shown:

```
Generated 25 entries:
[{'age': 19,
  'medical_risk': 'High',
  'name': 'Jeffrey Ross DWM',
  'salary': 70322,
  'zip_code': '33975'},
 {'age': 34,
  'medical_risk': 'High',
  'name': 'Stephanie Roberts',
  'salary': 111415,
  'zip_code': '10971'},
 {'age': 47,
  'medical_risk': 'High',
  'name': 'Chad Hoffman',
  'salary': 39614,
  'zip_code': '12267'}]

Dataset saved to 'employee_data.csv'
```

On the right side of the VS Code interface, there is a 'CHAT' panel with a 'Build with Agent' button and a text box for generating agent instructions.

1. Data Generation

Shows: Your `lab2.py` code and terminal output displaying "Generated 25 entries."

- This screenshot shows the generation of the `raw_data` list containing **25 unique entries**. Each entry includes necessary attributes such as **name**, **age**, **zip code**, **salary**, and **medical risk**. It also confirms the dataset has been exported to `'employee_data.csv'`.

The image shows a Visual Studio Code editor window with a file explorer on the left and a terminal at the bottom. The file explorer shows a project named 'LAB2_INFOE3_FRANCISCO_MANULAT_LABANON_CORTEJO_3C' containing files: 'employee_data.csv', 'lab2.py', 'swapped_employee_data.csv', and 'swapping_lab2.py'. The 'swapping_lab2.py' file is open in the editor, showing Python code that reads 'employee_data.csv', calculates pre-swap metrics (Total Payroll and Average Salary), and displays a sample of the original data before swapping. The terminal shows the command 'python swapping_lab2.py' being executed, followed by the output of the script, which includes the pre-swap metrics and a sample of the original data.

```
1 import pandas as pd
2 import random
3
4 # Load the data generated from the first script
5 df = pd.read_csv('employee_data.csv')
6 raw_data = df.to_dict('records')
7
8 # 1. Calculate Pre-Swap Metrics [cite: 247]
9 original_salaries = [record['salary'] for record in raw_data]
10 pre_swap_total = sum(original_salaries)
11 pre_swap_avg = pre_swap_total / len(original_salaries)
12
13 print("-" * 60)
14 print("      PRE-SWAP METRICS")
15 print(f"Total Payroll: ${pre_swap_total:,.2f}")
16 print(f"Average Salary: ${pre_swap_avg:,.2f}")
17 print("-" * 60)
18
19 # --- STEP A: DISPLAY SAMPLE BEFORE SWAPPING [cite: 192-194] ---
20 print("\nSAMPLE OF ORIGINAL DATA (Before Swap):")
21 for i in range(min(3, len(raw_data))):
22     name = raw_data[i]['name']
23     salary = raw_data[i]['salary']
24     print(f"    {name} - ${salary:,.2f}")
25 print("-" * 60)
26
27 # 2. Safe Shuffle Logic [cite: 261, 264-267]
```

PS D:\INFOE3 TASKS\Lab2-Infoe3\Lab2_INFOE3_Francisco_Manulat_Labanon_Cortejo_3C> python swapping_lab2.py

Chad Hoffman - \$131,356.00

✓ Swapped dataset saved to 'swapped_employee_data.csv'

PS D:\INFOE3 TASKS\Lab2-Infoe3\Lab2_INFOE3_Francisco_Manulat_Labanon_Cortejo_3C> python swapping_lab2.py

PRE-SWAP METRICS

Total Payroll: \$2,153,065.00

Average Salary: \$86,122.60

SAMPLE OF ORIGINAL DATA (Before Swap):

Jeffrey Ross DVM - \$70,322.00

Stephanie Roberts - \$111,415.00

Chad Hoffman - \$39,614.00

2. Pre-Swap Metrics

- This image displays the '**Pre-Swap**' calculations for the dataset. It establishes the baseline **Total Payroll** and **Average Salary**, which are needed to verify statistical accuracy throughout the swapping process. By calculating these values first, we ensure that the anonymized data remains functionally identical to the original for auditing purposes."

The screenshot shows a VS Code editor with a project named 'Lab2_INFOE3_Francisco_Manulat_Labanon_Cortejo_3C'. The Explorer panel on the left shows files: 'employee_data.csv', 'lab2.py', 'swapped_employee_data.csv', and 'swapping_lab2.py'. The main editor displays the 'swapping_lab2.py' script, which uses pandas to read 'employee_data.csv', calculate pre-swap metrics (Total Payroll: \$2,153,865.00, Average Salary: \$86,122.60), and save the swapped data to 'swapped_employee_data.csv'. The terminal at the bottom shows the command 'python swapping_lab2.py' being executed, with output confirming the metrics remain identical after the swap. A sample of original data (Jeffrey Ross DVM - \$70,322.00, Stephanie Roberts - \$111,415.00, Chad Hoffman - \$39,614.00) and swapped data (Jeffrey Ross DVM - \$54,024.00, Stephanie Roberts - \$131,356.00, Chad Hoffman - \$42,406.00) is also displayed. The status bar at the bottom indicates 'Ln 71, Col 66' and 'Python'.

```
1 import pandas as pd
2 import random
3
4 # Load the data generated from the first script
5 df = pd.read_csv('employee_data.csv')
6 raw_data = df.to_dict('records')
7
8 # 1. Calculate Pre-Swap Metrics [cite: 247]
9 original_salaries = [record['salary'] for record in raw_data]
10 pre_swap_total = sum(original_salaries)
11 pre_swap_avg = pre_swap_total / len(original_salaries)
12
13 print("-" * 60)
14 print("PRE-SWAP METRICS")
15 print(f"Total Payroll: ${pre_swap_total:,.2f}")
16 print(f"Average Salary: ${pre_swap_avg:,.2f}")
```

PS D:\INFOE3 TASKS\Lab2-Infoe3\Lab2_INFOE3_Francisco_Manulat_Labanon_Cortejo_3C> python swapping_lab2.py

PS D:\INFOE3 TASKS\Lab2-Infoe3\Lab2_INFOE3_Francisco_Manulat_Labanon_Cortejo_3C> python swapping_lab2.py

PRE-SWAP METRICS
Total Payroll: \$2,153,865.00
Average Salary: \$86,122.60

SAMPLE OF ORIGINAL DATA (Before Swap):
Jeffrey Ross DVM - \$70,322.00
Stephanie Roberts - \$111,415.00
Chad Hoffman - \$39,614.00

POST-SWAP METRICS
Total Payroll: \$2,153,865.00
Average Salary: \$86,122.60

SAMPLE OF SWAPPED DATA (After Swap):
Jeffrey Ross DVM - \$54,024.00
Stephanie Roberts - \$131,356.00
Chad Hoffman - \$42,406.00

✓ Swapped dataset saved to 'swapped_employee_data.csv'

PS D:\INFOE3 TASKS\Lab2-Infoe3\Lab2_INFOE3_Francisco_Manulat_Labanon_Cortejo_3C>

3. Data Swapping & Final Verification

- This screenshot displays the final execution of the data swapping process. It confirms that the salaries have been successfully redistributed among employees to protect individual privacy, while the **Total Payroll** and **Average Salary** remain statistically identical to the original dataset. The final anonymized file has been exported as **swapped_employee_data.csv**.